

CS264 CA 5

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1 Instructions

Please code up your solutions in cpp file(s) and upload your cpp files on moodle. Only C++ source code (**.cpp files or .h files**) will get accepted and any other format will be desk rejected immediately. Test cases are not mandatory and are for you own testing purpose. You must submit your source files before **17:00**.

2 Questions

Design a simple module management system. Figure 1 shows its UML class diagram. You may design any methods (including constructors) that can help you to implement the following features. For each class, you should put class definition in a .h file and class implementation in a .cpp file.

Key Features.

Each module (Module class) has a name, some credits (you can set whatever credits you like, by default it is 2.) and number of students who are registered for this module. You should allow a user to check if two modules are identical or not. Two modules are identical if they have the same name and credits. Implement this feature using operator overloading for ==.

Each Programme (Programme class) consists of a set of modules. The number modules of a programme can have is capped by a positive integer (this integer is a variable not a constant). You should only allow users to add a module if it does not exceed its capacity. Implement this feature using a pointer that points to an array of modules, and this must be done using dynamic memory management.

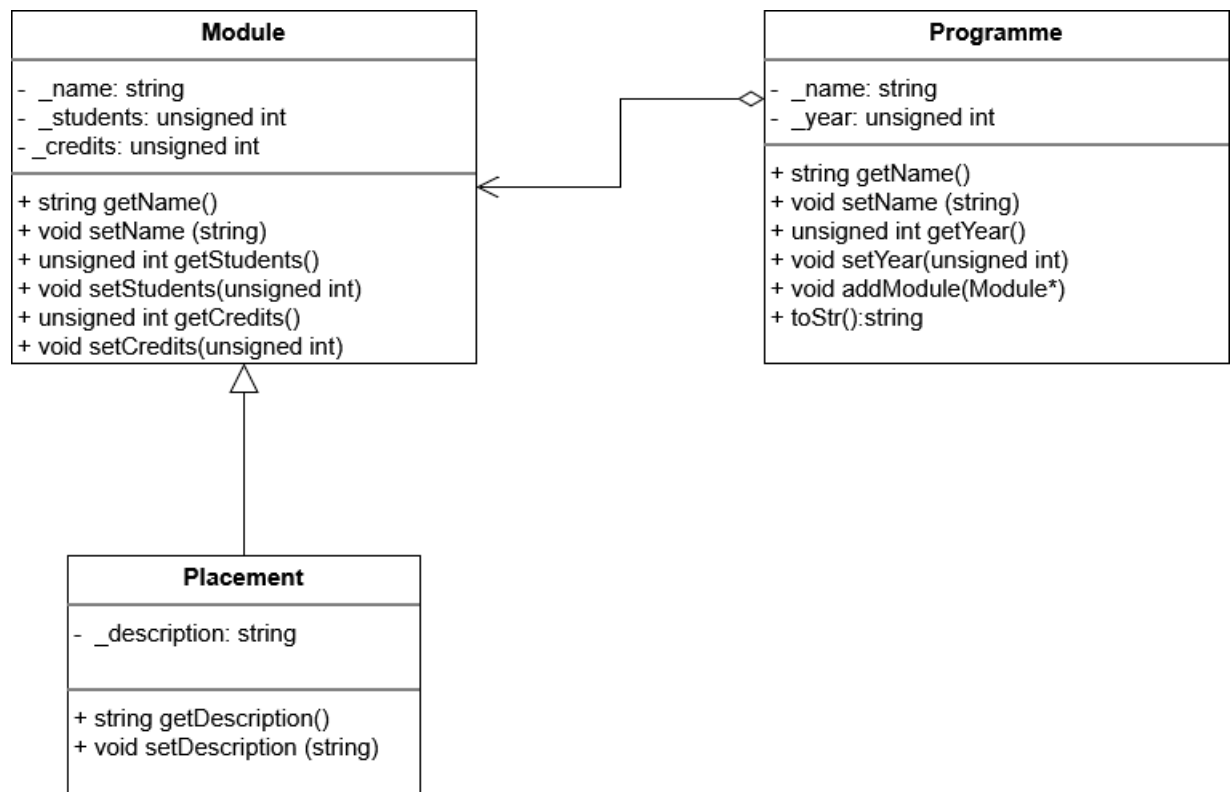


Figure 1: The UML class diagram for a simple module management system.

Each programme has a string representation (*toStr()* method in Figure 1): a string contains each module's name, credits and number of students studying this module.

A placement module is a special type of module that has some description only (any description is fine) and it is 15 credits.

3 Test Cases

Create at least two modules (CS264 and CS357) to test your design, and both modules have 5 credits. CS264 has about 181 students and CS357 has about 155 students. Add both modules into a programme called CSSE (3rd year) and print its string representation.